Best Practices in Teaching and Learning

As higher education undergoes a transformative shift with the integration of generative artificial intelligence (AI), it becomes imperative to establish and adhere to a set of best practices that maximize the benefits while mitigating potential drawbacks. This report provides a compilation of exemplary use cases, each illustrating the applicable principles and practical applications of generative AI within the realms of teaching and learning. These use cases not only embody foundational principles—such as Generative AI Literacy, Academic Integrity, and Inclusion, Equity, and Access—but also serve as a guide for instructors and learners to navigate the complexities of AI integration in teaching and learning processes.

The intent behind outlining these best practices is not to be exhaustive, but to provide a practical framework that instructors can utilize to incorporate generative AI tools in a manner that enhances learning outcomes, ensures ethical use, and promotes a balanced human-AI collaboration. From crafting personalized learning experiences to automating assessment and feedback, these practices showcase the diverse capabilities of generative AI in enriching the educational landscape.

In presenting these use cases, we aim to equip instructors and students with the knowledge and tools necessary to effectively leverage AI technologies, thereby fostering an educational environment where technology complements traditional teaching methodologies and supports the development of a future-ready learner. Through this exploration, we aspire to catalyze a broader understanding and acceptance of generative AI as a pivotal element in the evolution of educational practices.

Guided by the ambition to foster a tech-savvy and future-ready Illinois community, we recommend practical applications and actionable strategies that can bring these aspirations to fruition. The journey from conceptual understanding to practical application is pivotal, as it transforms abstract knowledge into tangible skills and insights. The desire for this transformation leads us to the recommended and crucial approach of facilitating interactive workshops, creating an environment where the theoretical dimensions of generative AI in teaching and learning crystallize through hands-on exploration and expert-led dialogue. These workshops are not just learning sessions but catalysts for change, designed to immerse both instructors and students in the practical realities of AI, ensuring they emerge not only as informed participants in the digital age but as proactive contributors to the ongoing narrative of educational evolution.

These interactive workshops should be dedicated to encouraging the ethical and practical aspects of AI in education, aiming to foster a community well-versed in AI literacy. Workshops should focus on interactive learning, customization to meet diverse needs, and providing resources for continued education outside the classroom. Additionally, offering recognition for participation would further encourage instructors' engagement and acknowledge the importance of staying informed in the rapidly evolving field of AI.Workshops and other engagements by various colleges and units are currently being held. Some of the efforts include GenAI sessions for instructors are being delivered by the Center for Innovation in Teaching and Learning and the Library. There is an open Illinois community around Generative AI in Education in Microsoft Teams. There is a partnership between Education, Business, Siebel Center for Design, and the Generative AI Solutions Hub that hosts monthly hybrid sharing sessions around the Gen AI use cases the Illinois community is utilizing titled GenAI Dialogs. There are apps being developed on campus around Gen AI and education including Illinois.Chat and Arist AI. These are just a few of the Illinois efforts. However, a more organized campus-wide effort would help instructors and learners be more aware of the offerings, provide a more consistent message, and help fill in gaps or needs that are identified.

Principles

The foundational principles listed here ideally underlie all that Illinois does as an institution. Specific principles will be aligned with best practices and will be highlighted later in this report.

Generative AI Literacy

- Instructors and learners need to know how and when to use AI appropriately.
- Instructors and learners need to understand how generative AI works.

Academic Integrity

- Instructors and learners should use AI in ethical ways.

- Instructors and learners must acknowledge when and how they have used AI in learning.
- Learners are responsible for doing the work to learn.

Keeping Humans in an Active Role

- Instructors and learners should balance AI and Human work.
- Instructors and learners should have access to a personalized learning experience with AI.
- Instructors should facilitate learning with AI.

Complementing Creativity

- Instructors should consider roles for AI within assignments that develop learner metacognitive skills.
- Learners should use AI to build on their own original, creative ideas and experiences, rather than replace them.

Inclusion, Equity, and Access

- Instructors and learners should be aware of potential biases in their use of AI.
- Instructors and learners should have equitable access to generative AI tools and their needs should be accommodated.

Data Privacy and Security

- Instructors and learners must have their privacy and security protected.
- Instructors and learners should receive training on data privacy and security best practices.
- Instructors need to comply with data protection laws and security practices.

Scalability and Integration

- Instructors and learners will have increasing demands for the use of AI, so it will need to be scalable.
- Instructors and learners should have access to AI tools that are integrated into existing campus tools.

These best practices were prepared by the University of Illinois Urbana-Champaign GenAI Solutions Hub: Teaching and Learning Working Group. These best practices were created with the assistance of artificial intelligence (AI) tools, Chat GPT, Microsoft Co-pilot, and others. These tools were utilized to enhance the quality, accuracy, and efficiency of the content. All content was created with humans in the loop, ensuring oversight and human judgment throughout the process.

Best Practices

Notify Learners of Expectations for Using Generative AI to Support Learning

Use Generative AI for Assessment and Feedback

Create and Customize Content for Learning

Acknowledge AI Use and Cite AI-Generated Content

Use AI for Knowledge Management and Information Retrieval

Foster Human / AI Collaboration

Leverage Personalized Learning Support

Foster Critical Engagement and Evaluation

Promote Inclusivity When Using AI

Use AI to Supplement Instruction

Ensure Data Privacy for Learners and Instructors

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Office of the Chief Information Officer

Office of the CIO and partner sites

Boldly IT at Illinois

Office of the CIO

Cybersecurity

Digital Accessibility

Enterprise GenAI

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